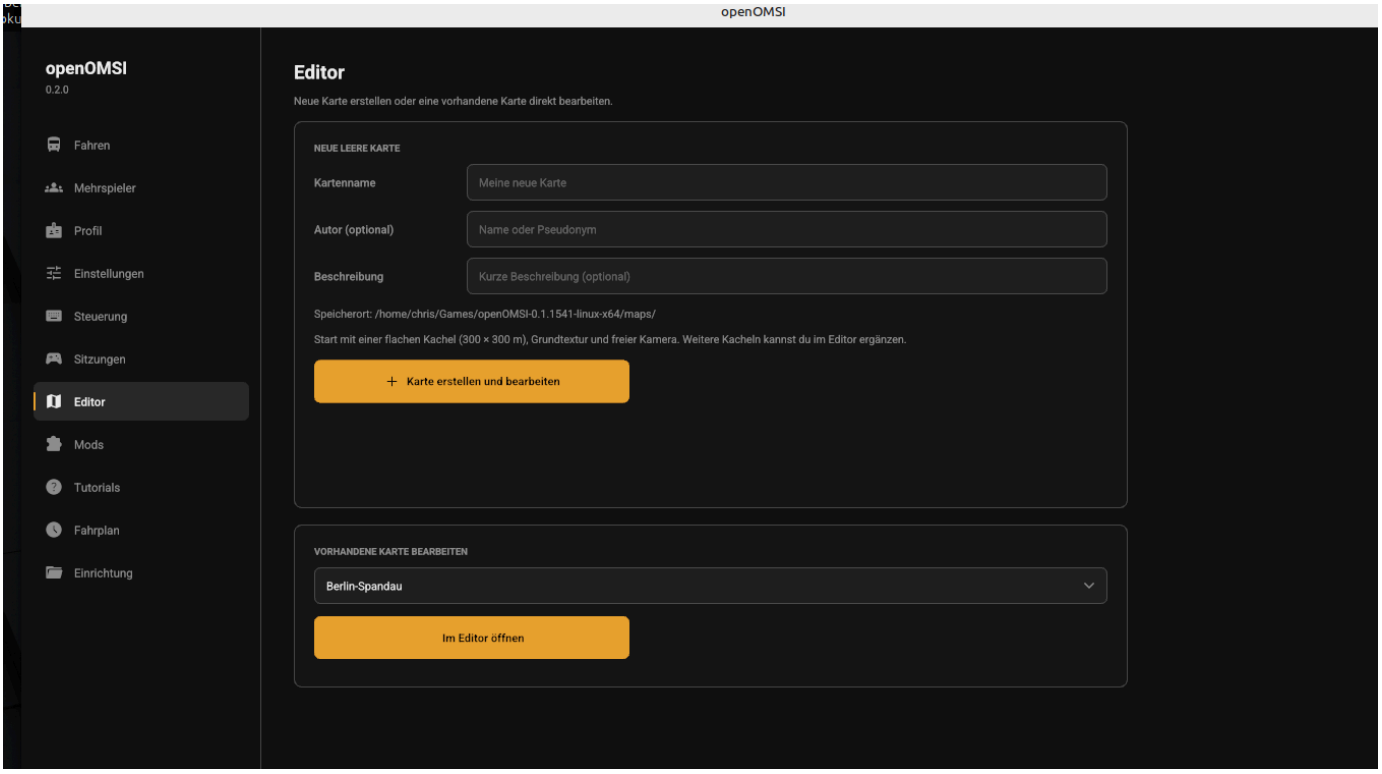
1 Start directly in the editor

In the launcher, choose **Editor** (map icon), below **Sessions** and above **Mods**.



Original screenshot: German launcher. The visible folder path and launcher version label are not installation instructions.

Create a new blank map

Kartenname (map name): required; also determines the folder name. Existing names are rejected.

Autor / Beschreibung (author / description): optional; saved in the map description.

Speicherort (destination): shown below the fields. Maps are created in the active openOMSI content folder’s maps directory, outside the original OMSI 2 installation.

Karte erstellen und bearbeiten (create and edit): creates one flat 300 × 300 m tile with a base texture and opens the editor. No bus or driving duty is needed.

2 Maps and camera position

Open an existing map

Select a map under **Vorhandene Karte bearbeiten**, then click **Im Editor öffnen**. End any running session first. The map opens with a free camera and the editor already active.

Start above the terrain

With no saved editor position, the standard is the same elevated view as on a new map: **35 m above the terrain** at the start location, looking **30° downward**. If the map’s initial location lies outside existing tiles, the nearest tile is used.

Return to the last editor location

Position, direction and zoom are remembered per map when you **save**, **switch off the editor** or **end the session normally**. The next direct editor start restores that location. Height is raised to at least 35 m above the current terrain when needed; higher saved positions are retained.

Save and reopen

Fly to a recognisable location, make a desired edit and press **Ctrl+S**. Close the session and reopen the same map through **Editor**. Check the edit and camera position. Other maps retain their own positions.

What the camera bookmark saves

Camera data is stored locally under **~/.openomsi/editor-views**, not in the map. It does not save unsaved map edits. A crash may lose the latest location that has not yet been recorded. Normal driving starts are unchanged.

Language and available tools

The editor does not yet fully follow in-game language changes; some labels remain German. The German and English follow the language selected in the game. Other languages are not fully translated yet. Junction and roundabout builders are included. New features are covered on pages 13-16 .

3 Open the editor during play

Edit the currently loaded map directly during gameplay.

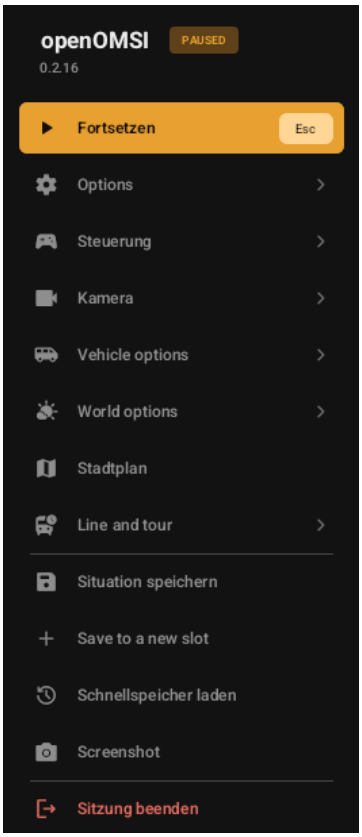
New: direct editor start and blank maps - see pages 1-2.

Press **F4** to switch to camera mode. Then press **Esc** to open the pause menu.

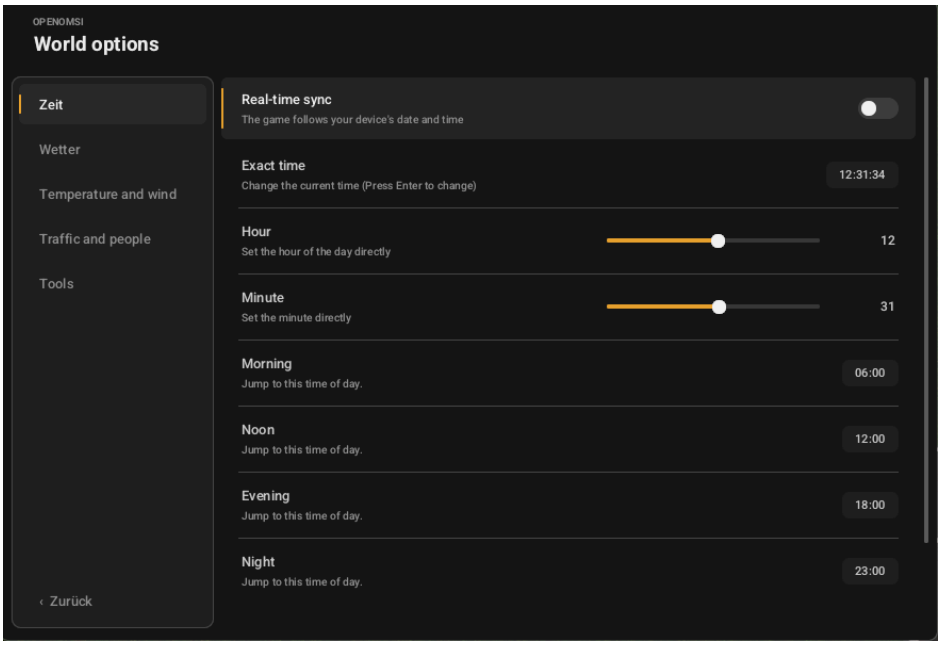
World options → Tools → Objekteditor → Open

This opens the Object Editor. Press T to switch to the Spline Editor for roads (page 10). Screenshots use German or mixed-language labels; this guide includes the relevant German names.

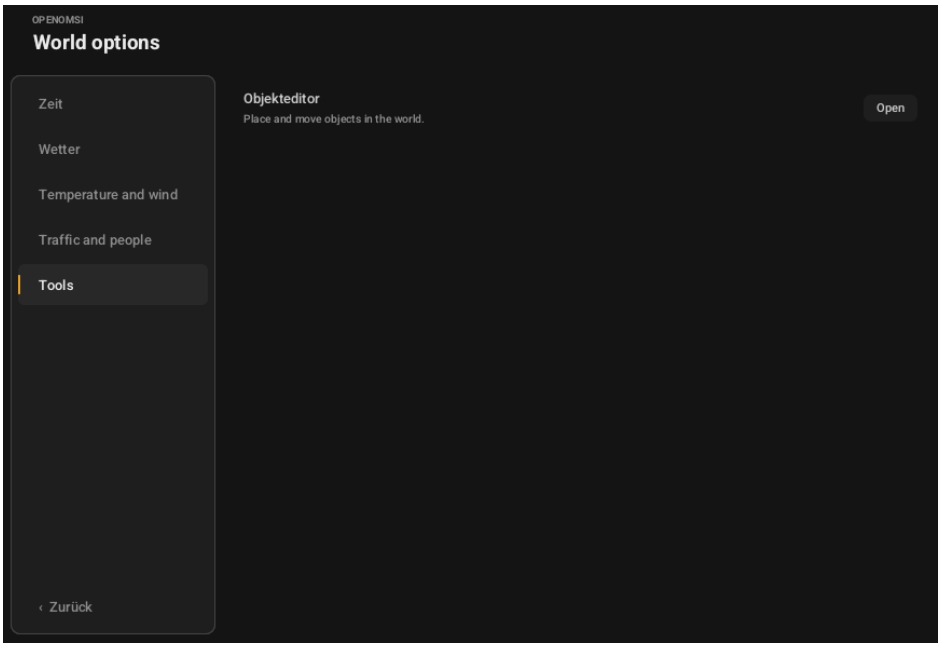
PAUSE MENU



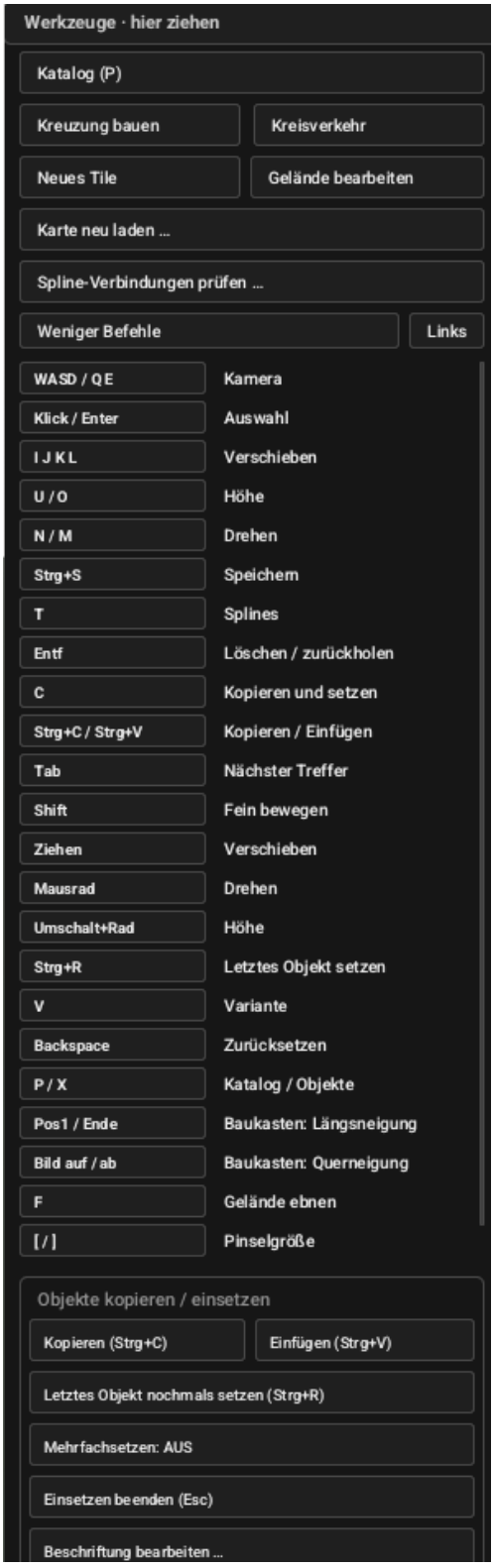
WORLD OPTIONS



TOOLS: OPEN THE OBJECT EDITOR



4 Interface and objects



Current screenshot with German labels; lower panel area is cropped.

Tool panel

Catalogue, Build junction and Roundabout open the object tools. Below are new tiles, terrain editing, map reload and spline connection checks.

Panel and commands

Drag the panel by its header. “All commands” expands the shortcuts; “Fewer commands” collapses them. Scroll to see more commands.

Selection and information

The information bar shows the editing mode and selected object. Click or Enter selects; Tab cycles through hits.

Copy and place

Ctrl+C copies and Ctrl+V starts placement. Ctrl+R uses the last placed object. Multi-place allows repeated placement; Esc stops placement.

Rotation and tilt

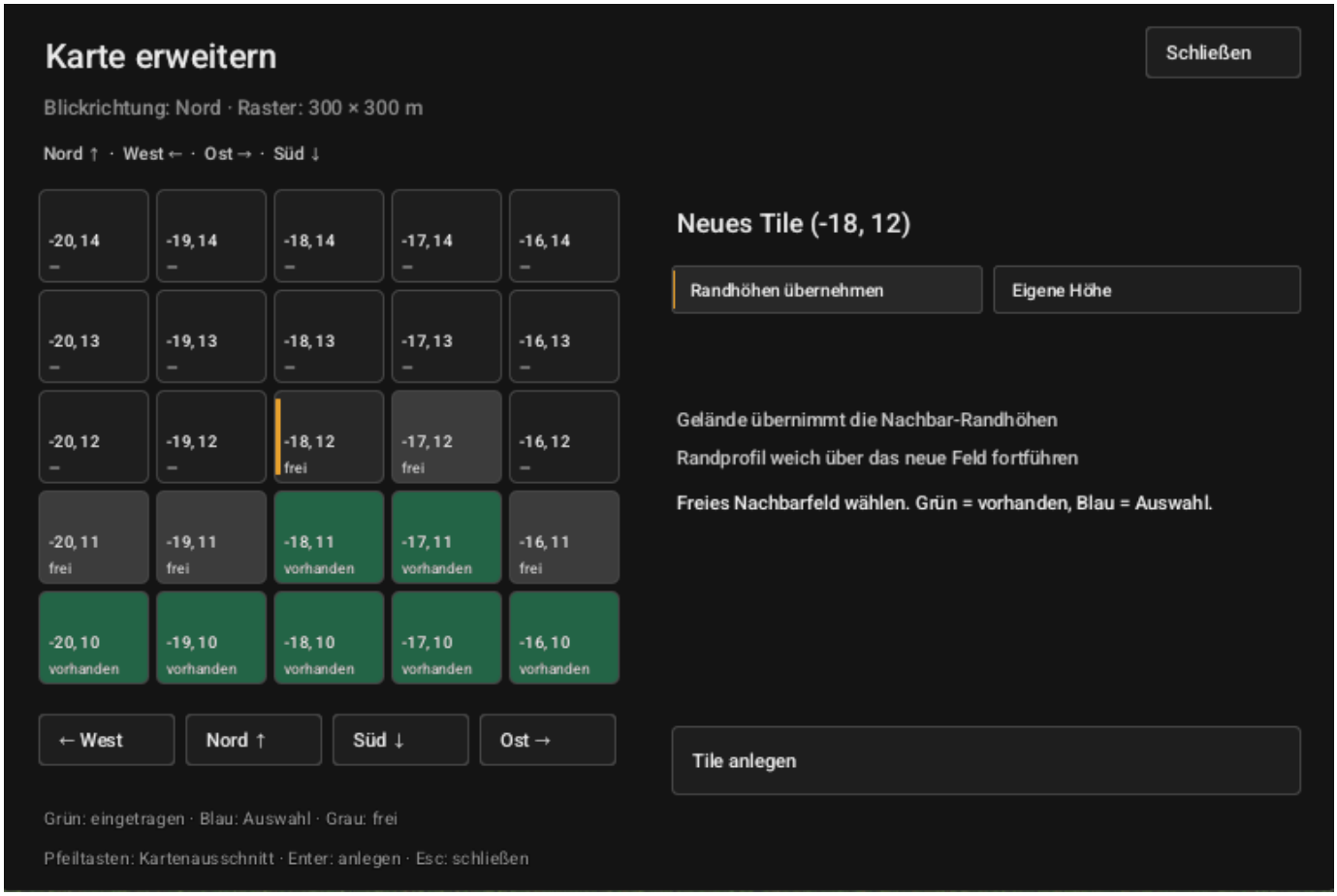
N/M rotates, U/O changes height. Home/End changes builder object pitch; Page Up/Down changes bank. Hold Shift for finer adjustments. See page 15.

Save

Ctrl+S saves the map. Save desired changes before reloading.

5 Adding map tiles

“Neues Tile” opens the map expansion window. Each tile measures 300 × 300 m.



Choose a free neighbouring tile

Green marks existing tiles; "frei" marks free positions. Your selection is dark grey with an orange side marker. The on-screen legend still says "blue". Each cell shows its coordinates. Use the direction buttons or arrow keys to move the map area.

Set the terrain height

“Randhöhen übernehmen” takes the edge heights of neighbouring tiles and smoothly extends their profiles across the new tile. **“Eigene Höhe”** lets you specify a height yourself.

Create the tile

Click **“Tile anlegen”** or press **Enter**. “Schließen” or **Esc** closes the window.

Choose a free position beside the existing map to add a connected area for building.

6 Editing the terrain

Change the ground shape or switch between heights and textures.



Height tools

Open “Gelände bearbeiten”. The “Höhen” tab provides tools to raise, lower, level and smooth the ground. “Texturen” switches to textures.

Brush settings

Radius: size of the affected area.
Stärke / Sek.: strength per second.
Weicher Rand: soft edge falloff.
Zielhöhe: target height for applicable tools.
Tile-Schritt: step size for whole tiles.

Pick a height or tile

“Höhe aufnehmen” samples an existing terrain height. “Tile auswählen” selects a map tile for the tile tools; its coordinates appear below.

Edit entire tiles

Raise or lower the tile, or set it to the target height. “Übergang zu Nachbarn” controls the transition width towards neighbouring tiles.

Paint and save

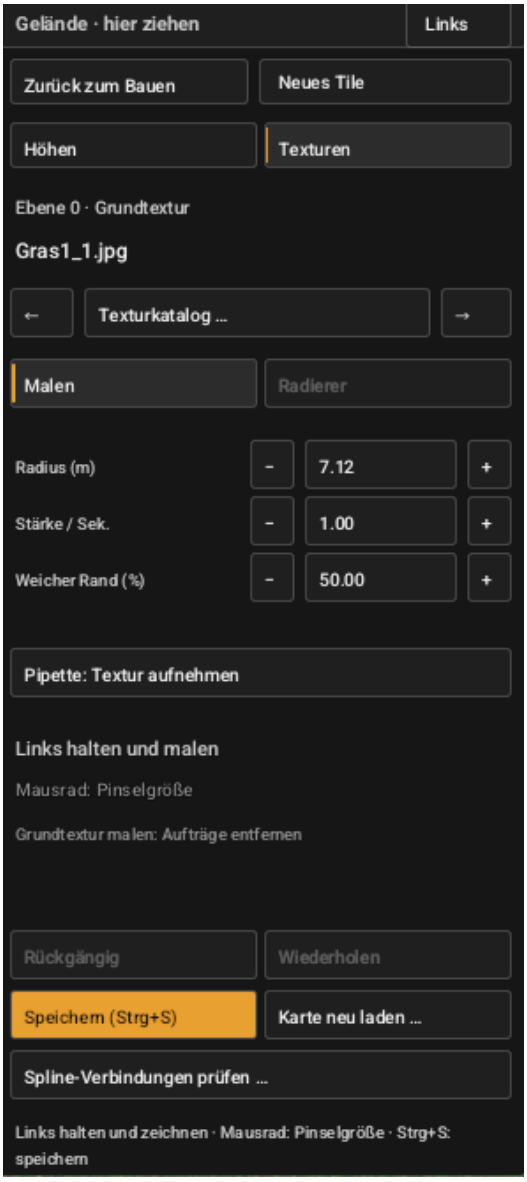
Hold the **left mouse button** to paint. Hold **Shift** for finer adjustments. “Rückgängig” and “Wiederholen” undo and redo changes. Save with **Ctrl+S**.

“Zurück zum Bauen” returns to building mode.

The "Texturen" tab is explained in detail on page 7.

7 Painting terrain textures

Open "Gelände bearbeiten", then select "Texturen".



Choose a texture and layer

The arrows switch terrain texture layers. "Texturkatalog ..." opens texture selection. The current layer and texture name appear above these controls.

Paint or erase

Select "Malen" and hold the left mouse button over the terrain. "Radierer" removes the selected overlay layer. It is disabled on layer 0: painting the base texture removes overlaid paint instead.

Brush settings

"Radius (m)" sets brush size; the mouse wheel also changes it. "Stärke / Sek." controls how quickly paint is applied or erased. "Weicher Rand (%)" controls the soft transition at the edge.

Pick an existing texture

Click "Pipette: Textur aufnehmen", then click the terrain. The editor selects the sampled terrain texture layer.

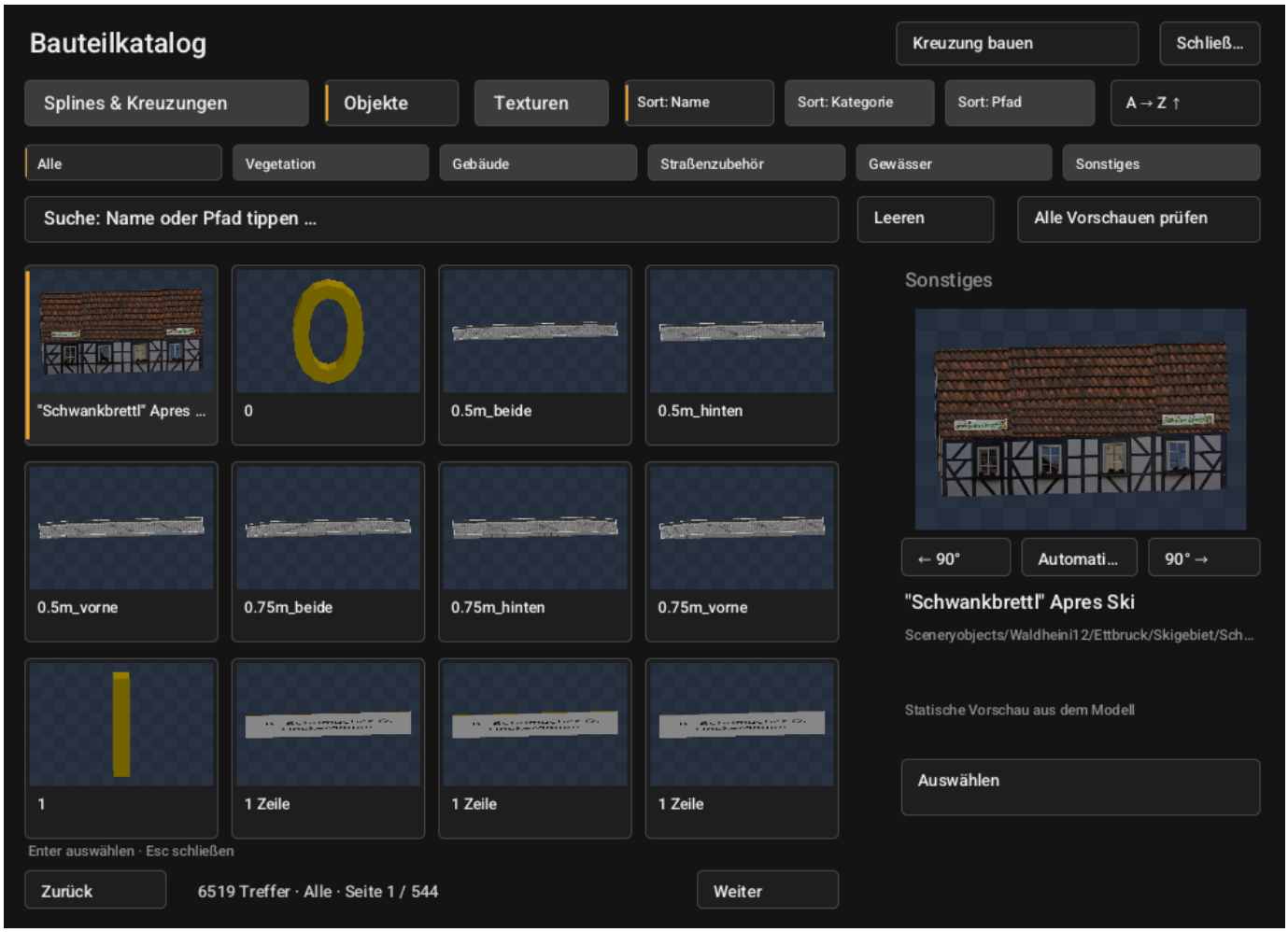
Undo and save

Use "Rückgängig" / "Wiederholen" to revise brush strokes. Save with "Speichern (Strg+S)". "Zurück zum Bauen" returns to the building tools.

Tip: begin with a small brush and low strength. Terrain textures change the ground surface; they do not select the asphalt texture of a spline or junction.

8 The asset catalogue

Open “Katalog (P)” in the tool panel, or press P.



Search and filter

The top tabs switch between **Splines & Junctions** (“Splines & Kreuzungen”), **Objects** (“Objekte”) and **Textures** (“Texturen”). Categories narrow the results. Search by name or file path; “Leeren” clears the search.

Preview and selection

Results appear as thumbnails on the left. Click an item to see a larger preview, its name and file path on the right. The buttons below the preview rotate it.

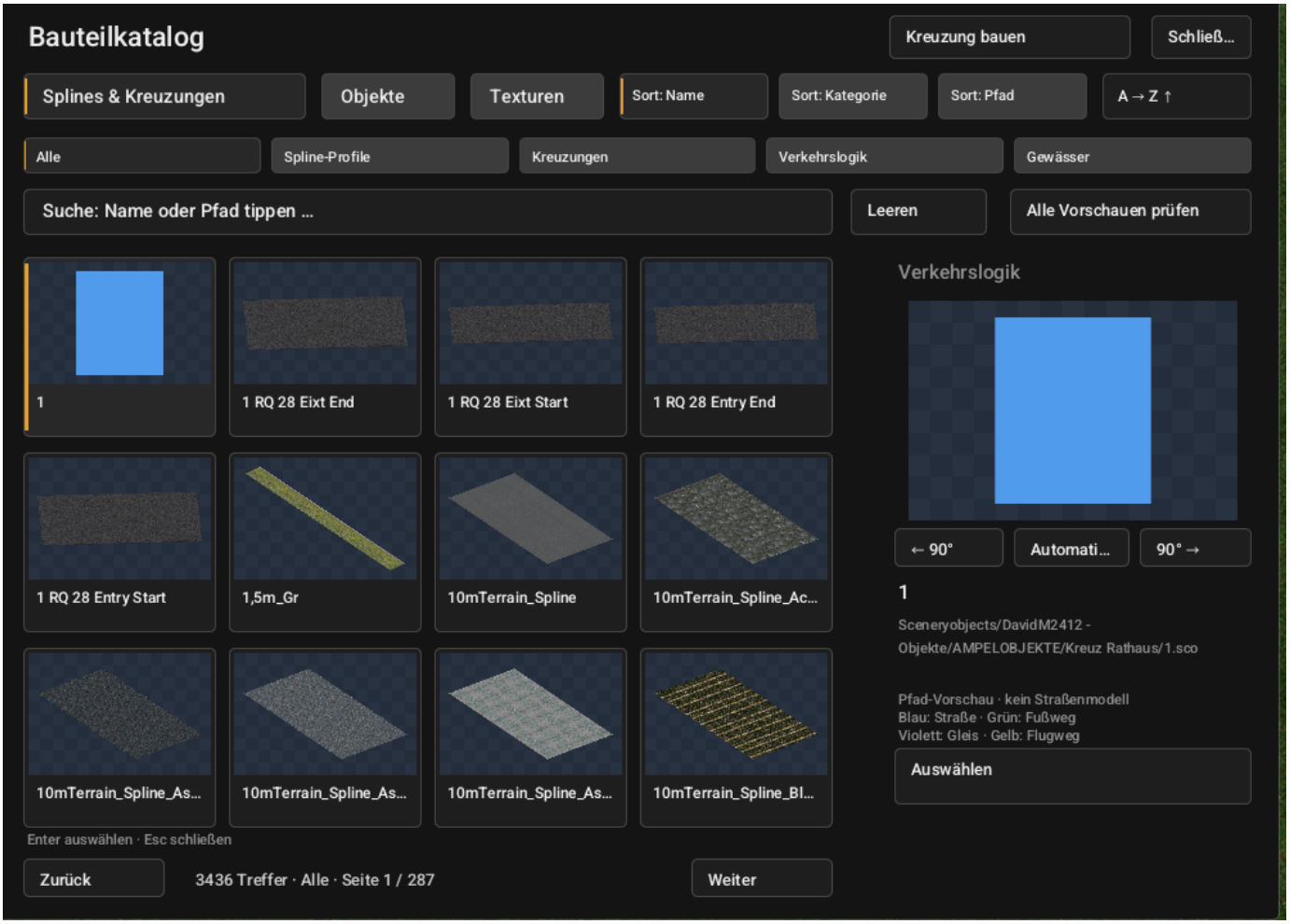
Select and browse

Click **“Auswählen”** (Select) or press **Enter** to use the highlighted item. “Zurück” and “Weiter” move between result pages. **Esc** closes the catalogue.

“Kreuzung bauen” opens the junction builder. “Alle Vorschauen prüfen” starts a check of the catalogue previews.

9 Splines in the catalogue

Use "Splines & Kreuzungen" to find road profiles and junction components.



Filter and search

Choose categories such as "Spline-Profile", "Kreuzungen" or "Verkehrslogik". Search by name or file path and use the sorting controls to narrow the list.

Understand the preview

A traffic-logic preview represents paths, not necessarily a visible road model. Blue means road, green footpath, violet rail and yellow flight path. Read the preview description on the right.

Use the selection

Click "Auswählen" or press Enter. "Kreuzung bauen" opens the junction builder; its texture controls are explained on page 12.

10 The Spline Editor

Select and edit roads, then add AI driving paths.



Current tool panel. Screenshot labels are in German.

Mode and selection

Press T in the Object Editor to enter spline mode. Click a road; Tab cycles through overlapping hits. Check the selected name and ID in the information bar.

Edit the road

+ / -: length; comma / period: curve; Home / End: gradient. C extends, G connects. Hold Shift for finer steps.

Road and terrain

F6 fits to ground. F8 smooths the edge; Shift+F8 smooths a wider area. Separate tools place objects and sidewalks along a road.

New: AI driving paths

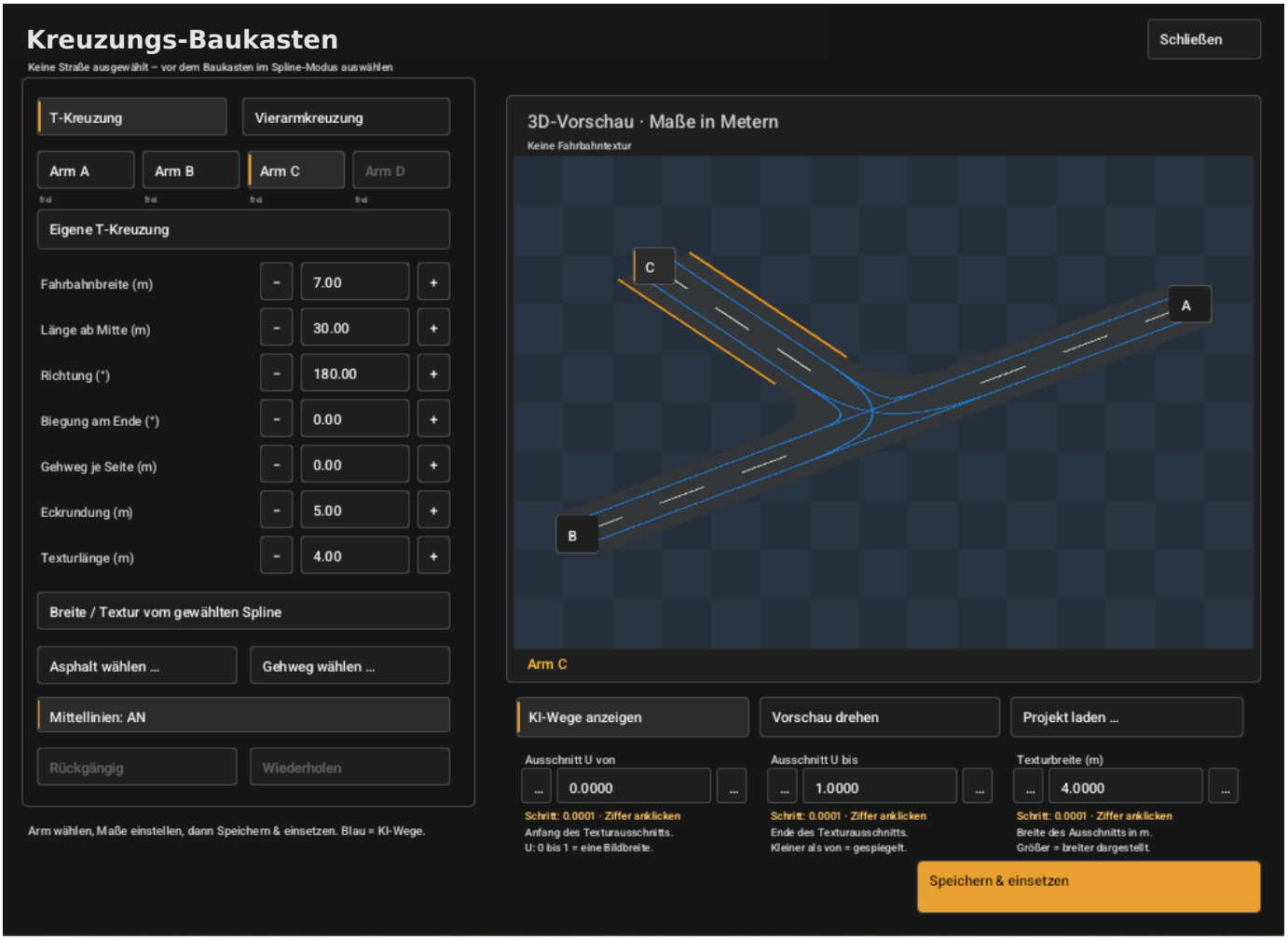
Open KI-Fahrwege entlang der Straße (AI paths along the road). KI-Pfade anzeigen / ausblenden shows or hides paths. Visible road geometry does not automatically provide an AI path.

Apply and save

Create AI paths: page 19. Drag points: page 20. Connections: page 21. Apply changes, save with Ctrl+S, then reload the map to update AI traffic.

11 Junction builder

Design a junction, save it and place it on the map.



Current preview with arm labels. German screenshot title edited to omit the version number.

Shape and arms

Choose a **T-junction** or **four-arm junction**. Select arm A/B/C/D; D is disabled for a T-junction. Set width, length, direction, bend and sidewalk width. Preview labels identify the arms; C is highlighted in orange here.

Textures and preview

Select asphalt and sidewalk textures. **Breite / Textur vom gewählten Spline** copies width and texture from a road selected beforehand. AI paths appear in blue; centre lines can be toggled.

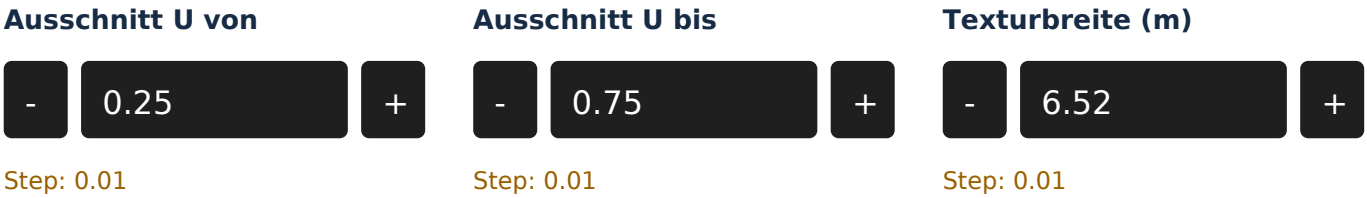
Save and continue editing

Load a saved project or use Undo/Redo to correct changes. **Save & place** prepares placement; then save the map with **Ctrl+S**. Texture crop: page 12; road connections: page 16; AI connections: page 21.

12 Adjusting junction textures

Editor 0.7.22-pre / openOMSI 0.2.27. Road texture settings for all junction arms.

Control diagram in the current style (not a screenshot)



Choose the step size

Click the value and type a number; comma and decimal point are accepted. Enter applies it. Plus/minus also applies a valid entry before changing it. The number of typed decimal places sets the step: 6.5 → 0.1; 6.52 → 0.01; 6 → 1. Up to four decimal places are supported.

Click a digit to select its place: in 6.52, clicking 6 selects steps of 1, clicking 5 selects 0.1, and clicking 2 selects 0.01. The selected digit is underlined. Use + or - to change the value. Each field remembers its own step.

What the fields do

Ausschnitt U von

Start of the horizontal texture crop. U = 0 is the left edge; U = 1 is one full image width.

Ausschnitt U bis

End of the crop. Example: 0.25 to 0.75 uses the middle half. A smaller end than start mirrors the crop; values outside 0 to 1 repeat the texture.

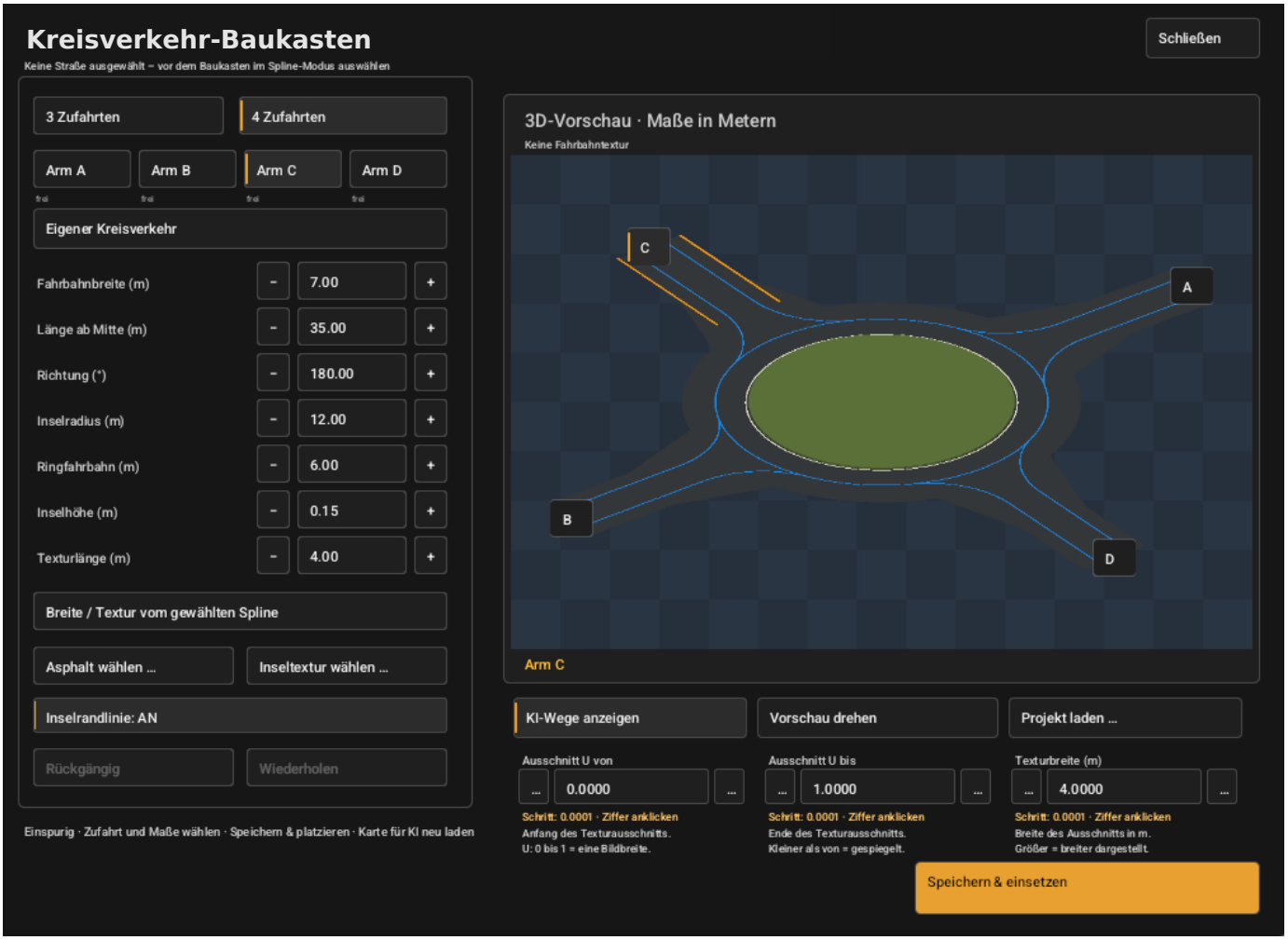
Texturbreite (m)

Width in metres covered by one repetition of the selected crop. Larger values stretch it wider; smaller values repeat it more often. The road width stays unchanged.

Limits: U from/to -16 to 16; the values must differ by at least 0.0001. Texture width: 0.25 to 100 m. Undo/Redo also covers these changes. Esc cancels a pending text entry; use Undo for changes already applied with +/-. Inspect the preview, then save or update the placed junction.

13 Roundabout builder

Choose approaches, identify them in the preview and place the roundabout.



Current preview with arm labels. German screenshot title edited to omit the version number.

Approaches and arm selection

Open **Roundabout** in the tool panel. Choose three or four approaches and edit arm A/B/C/D. The preview shows matching letters; selected arm C is highlighted in orange here.

Dimensions, textures and saving

Controls are explained on page 14. A previously selected road can supply width and texture. Use **Save & place**, place the object, then press **Ctrl+S**. Select a placed object and reopen the builder to edit it.

Check connections

Road connections: page 16. AI endpoints: page 21. After saving, reload the map so AI traffic uses the updated network.

14 Roundabout: fields and controls

3 / 4 approaches; Arm A-D	Choose the number of approaches and the arm to edit. Road width, length and direction apply to that arm.
Name	Name of your custom roundabout project.
Road width (m)	Width of the selected approach, not the circulating lane.
Length from centre (m)	Distance from the roundabout centre to the outer end of the selected approach.
Direction (°)	Orientation of the selected approach around the centre.
Island radius (m)	Radius of the central island. Larger values enlarge the island and roundabout.
Ring road width (m)	Width of the circulating roadway around the island.
Island height (m)	Height of the island above the roadway.
Texture length (m)	Length of one texture repeat along the roadway. Larger values stretch the pattern.
Width / texture from selected spline	Uses the previously selected road’s width and texture. Select a road in spline mode first.
Choose asphalt / island texture	Select textures for the roadway and central island.
Island edge line	Turns the visible island edge marking on or off.
Show AI paths / Rotate preview	Shows traffic paths or rotates the preview view. Rotate a placed object separately with N/M.
Load project; Undo / Redo	Load a saved builder project or undo/redo changes within the builder.
Crop U from / to; Texture width	Image crop and repeat width across the road. Click a digit and use +/-; see page 12 for details.

Input: click a numeric value, type and press Enter. The +/- buttons change dimensions. Limits: single circulating lane, no turbo roundabouts; direction follows the map’s left-/right-hand traffic. Check AI traffic after reloading.

15 Rotation, height and tilt

Select a placed object in **object mode**. Tilt controls support builder junctions and roundabouts. Keep the companion **junction.junction.json** beside the object.

Keys	Action	Normal / Shift
N / M	Rotate	5° / 0.5°
U / O	Lower / Raise	0.5 m / 0.05 m
Home / End	Pitch + / -	0.5° / 0.05°
Page Up / Down	Bank + / -	0.5° / 0.05°
I / J / K / L	Move	0.5 m / 0.05 m
Delete	Delete / Restore	
Ctrl+S	Save map	

Check the angles

The information bar shows pitch and bank. Axes follow the object, not the camera. Copying and placement preserve the angles. Each angle is limited to -89°…+89°; road connections may reach geometric limits earlier.

Backspace and saving

For connected builder objects, Backspace restores the combined pose and road edits. Without connected approaches it first resets changed tilt. Restore deleted objects with Delete first. Save and reload the map to refresh traffic paths.

Spline mode: comma/period still adjust curvature and Home/End adjust gradient. Keyboard rotation in object mode uses N/M only.

16 Connect roads and follow changes

Applies to builder junctions and roundabouts. A road that merely touches the object is **not yet an explicit connection**.

1. Select the approach road in spline mode.

2. Switch to object mode and select the placed builder object.

3. Open the appropriate builder and choose approach A/B/C/D.

4. Choose “Connect road to selected arm” to prepare the preview.

5. Check the preview and update the placed object to apply it. Closing the builder discards the preparation.

6. Save with Ctrl+S and reload the map to refresh traffic paths.

What follows automatically

Explicitly connected approaches follow moving, rotating, raising and tilting the object. Their far endpoints stay fixed. New connections to tilted objects are also supported. Backspace restores combined pose and approach edits together.

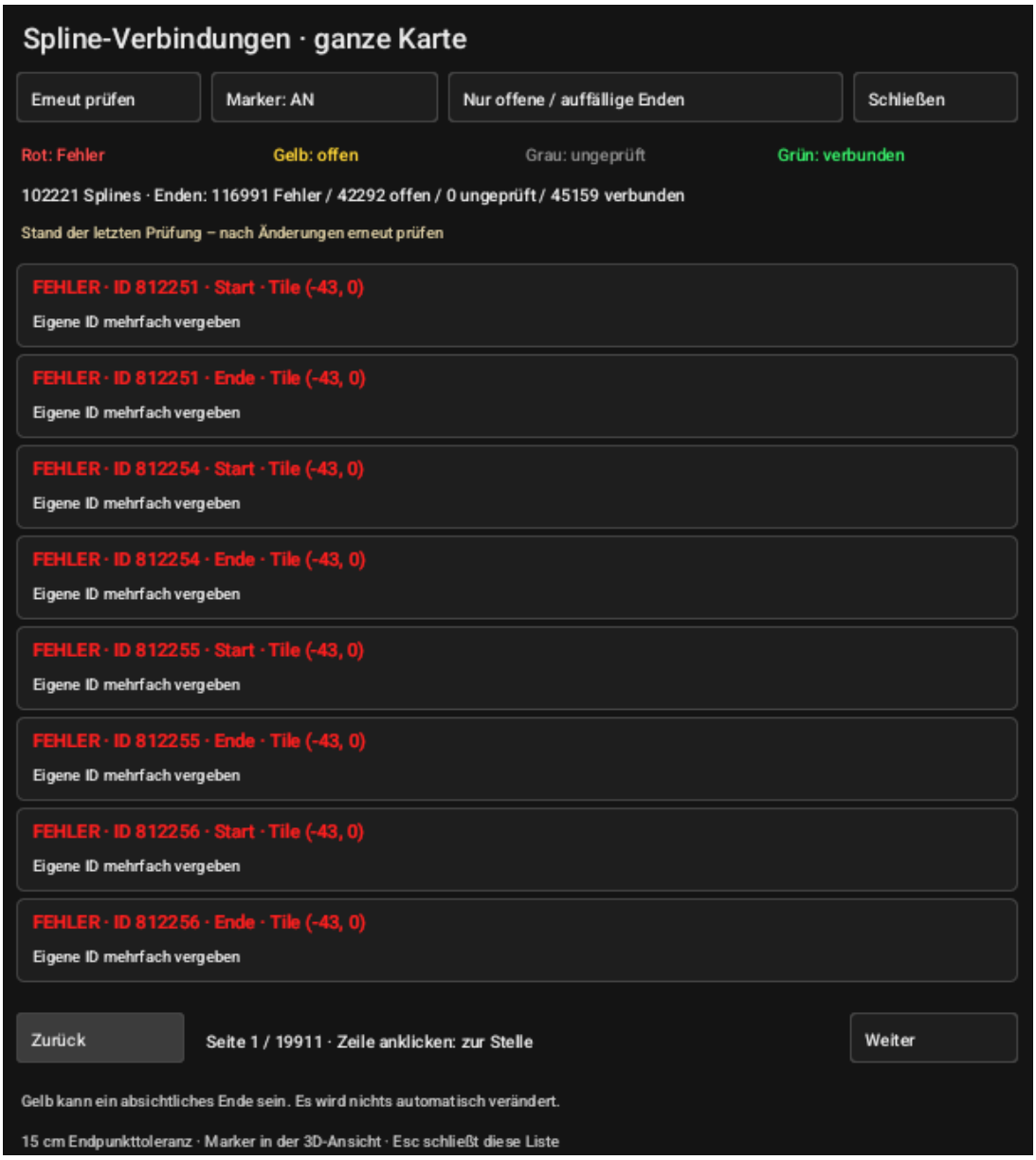
If a connection cannot be fitted

All approaches are checked before applying a pose change. If one fit fails, object and road geometry remain unchanged. Smaller changes, longer approaches or a matching transition spline may help. Repair an old broken link by selecting its road and reconnecting the correct arm.

Limits: standard OMSI maps; at most 50 m gap and 30° direction mismatch, compatible profiles and sufficient transition length. Reset protects separately edited roads. Non-editable Chrono or old-format roads are not changed.

17 Checking spline connections

Find open ends and suspicious connections across the whole map.



Red: an error, such as a duplicate ID. **Yellow:** an open end, which may be intentional. **Grey:** unchecked. **Green:** recognised as connected.

Each entry lists the spline ID, start or end, tile coordinates and the finding. **Click an entry to jump to that location.** “Zurück” and “Weiter” move between result pages.

“Erneut prüfen” runs the check again after edits. **“Marker: AN/AUS”** toggles markers in the 3D view. The filter “Nur offene / auffällige Enden” limits results to open or suspicious ends.

Endpoint tolerance: **15 cm**. The check does not change anything automatically. “Schließen” or **Esc** closes the list.

18 Version, startup and updates

Editor 0.7.22-pre / openOMSI 0.2.27 / 10.10.2026

What this edition adds

Spline tools: page 10. AI paths, vehicle permissions and freeform points: pages 19-20. Connections, opposing traffic and arm labels: page 21. Troubleshooting and diagnostics: page 22.

Install on Linux

Close the editor and game, back up maps and extract the new source package into its own folder. Run bash INSTALLIEREN.sh there. The installer checks and compiles the program. Start the new installation only after FERTIG (finished). Existing target folders are not overwritten.

Normal startup

From the installed editor folder:

```
bash start-editor.sh "/path/to/content-folder"
```

The content folder must be configured. OMSI 2 game content is not included. The source folder and the installed editor folder are separate locations.

Optional diagnostics

start-editor-diagnose.sh enables extra timing logs for that run. A subsequent normal launch with start-editor.sh disables them again. The smoother-dragging change stays active.

Testing status

The user confirmed dragging without freezes in 0.7.20 on Linux. This is not a complete test of every map or a Windows test. One map update still took about 0.49 seconds in a recorded session.

Appearance and screenshots

The bright system cursor remains; mouse steering is suspended in the editor. Older screenshots may show earlier version numbers. The text in this edition describes the current version. German and English are supported; other translations remain incomplete.

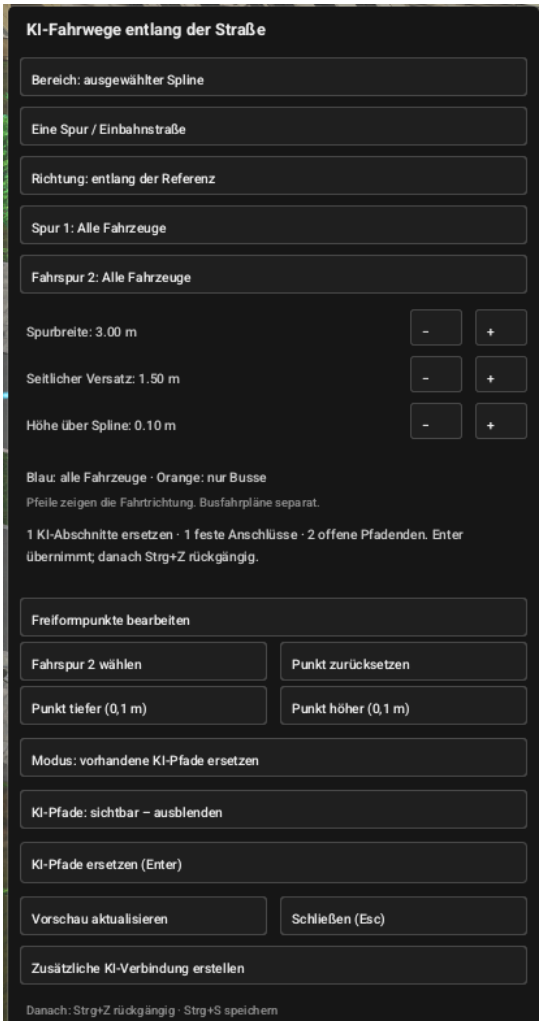
Downloads and releases

Updating source code does not change an existing public download. A new release package must be published separately. Freeform AI paths require this modified runtime; see page 21.

New in 0.7.22-pre: create an additional AI connection - step-by-step guide on page 23.

19 Create AI driving paths

First select the actual road, not an island, kerb or sidewalk.



Scope (Bereich)

Ausgewählter Spline edits one section. Verbundene Splinekette follows connected reference splines; junctions or gaps may limit the chain.

Lane count and direction

Cycle between opposing lanes, one one-way lane and two lanes in the same direction. Arrows show travel direction. Separately created one-way splines are edited individually.

Vehicles

Choose Alle Fahrzeuge (all vehicles) or Nur Busse (buses only) per lane. Bus schedules are configured separately; this permission does not create a bus service.

Width, offset and height

Lane width and lateral offset define the starting layout. Height above spline lifts the path relative to the road reference. Check that it follows the lane centre. This does not reshape the visible road.

Add missing paths or replace existing ones

Fehlende KI-Pfade ergänzen adds missing paths and keeps existing ones. Vorhandene KI-Pfade ersetzen updates editor-generated paths. Do not add another path when you intend to correct an existing one.

Important screenshot detail

This screenshot is set to Eine Spur / Einbahnstraße (one lane / one-way). The still-visible Fahrspur 2 wählen button does not create a second lane. A second editable lane requires a two-lane setting.

20 Shape AI paths with the mouse

Freeform editing works on one selected section at a time.

1 Open the existing path

Select the road or generated AI spline and open KI-Fahrwege entlang der Straße. Choose replacement mode when correcting an existing path.

2 Enable freeform editing

Click Freiformpunkte bearbeiten. The tool switches to the selected section. Each lane starts with nine green points, including its start and end.

3 Choose the lane

With two lanes, use Fahrspur 1/2 wählen to select the desired lane. The button names the lane you can switch to. Dragging does not move the other lane.

4 Drag points

Hold the left mouse button on a green point and drag. The selected point turns yellow: this means selected, not connected. Intermediate points shape a smooth curve. Keep the path in the centre of its traffic lane.

5 Adjust height or reset a point

Mouse dragging moves points horizontally. Punkt höher / tiefer changes height in 0.1 m steps. Punkt zurücksetzen resets that point's displacement. Displacement is limited to 50 m in each local direction.

6 Apply, save and reload

Apply with KI-Pfade anlegen / ersetzen or Enter. Ctrl+Z then undoes the applied change. Save with Ctrl+S and reload the map so running AI traffic uses the updated network. Closing or Esc discards the unapplied preview.

Longer routes and smooth dragging

Edit connected sections individually and join their endpoints. The chain function does not overwrite individual freeform shapes as one uniform layout. Since 0.7.20, mouse movements are combined per frame; release uses the latest position.

21 Connections and opposing traffic

Paths touching visually are not yet an explicitly selected connection.

Connect endpoints explicitly

In freeform mode, select the green start or end point. Matching opposite endpoints within 50 m appear purple. A path end connects to another path start, and vice versa. Drag the selected point onto the desired purple target and release; it snaps within 1.5 m.

Check the result

The panel counts feste Anschlüsse (fixed connections) and open ends. Check the assignment and arrow direction, then apply and save. Moving that endpoint freely again removes its fixed assignment. Nearly overlapping targets may be ambiguous.

Separate straight and right-turn lanes

Connect the straight-ahead lane to the crossing's appropriate straight path and the turning lane to its right-turn path. Merely straightening the visible line does not select the connection. Use two same-direction lanes for adjacent lanes travelling the same way.

Route around a central island

Keep each direction in the centre of its own lane. With right-hand traffic, pass to the right of the island in the direction of travel. Edit opposing lanes separately. With two separate one-way splines, the opposite direction is lane 1 of its own spline.

Arm labels A/B/C/D in the preview

The junction and roundabout builders label their arms in the preview. Match the label to the selected approach. This assists the road connections in chapter 16; AI endpoints are connected in the AI-path tool.

Limits and compatibility

Original paths in standard spline profiles or crossing objects are not overwritten. Custom paths can connect to suitable original endpoints. Free bridges across a crossing do not create traffic-light control or priority rules. Freeform points and fixed connections require this modified openOMSI runtime; an unmodified upstream version does not interpret these extensions.

New in 0.7.22-pre: create an additional AI connection - step-by-step guide on page 23.

22 AI paths: help and diagnostics

Check the selection and preview before applying and saving.

No paths or control points visible

Show AI paths. Enable freeform mode to see green control points. Background loading may take a moment after enabling the display. Check the information bar: did you select the road or the traffic island? An error can prevent a valid preview.

Cannot select lane 2

Eine Spur / Einbahnstraße has only one editable lane, although the second button may still be visible. If the opposite direction belongs to another AI spline, select it separately. Avoid creating a duplicate path over an existing one.

Curve too tight for this lane offset

A section does not fit the configured lateral offset. In chain mode, another section may be responsible. Select the correct road individually and check width, offset and curvature; do not use the narrow island edge as the reference.

Yellow point will not snap

Yellow means selected. Hold the left mouse button and drag the endpoint onto the matching purple target, then release. Check direction, distance and the 50 m displacement limit. Interior control points are not connection endpoints.

Another prolonged freeze?

Launch with start-editor-diagnose.sh. Note the time and action: dragging, releasing, applying or moving the camera. Save game.log before the next launch. EDITOR-DIAG reports slow operations; 0.7.20 also reports excessive waiting of combined mouse movements.

What has been observed

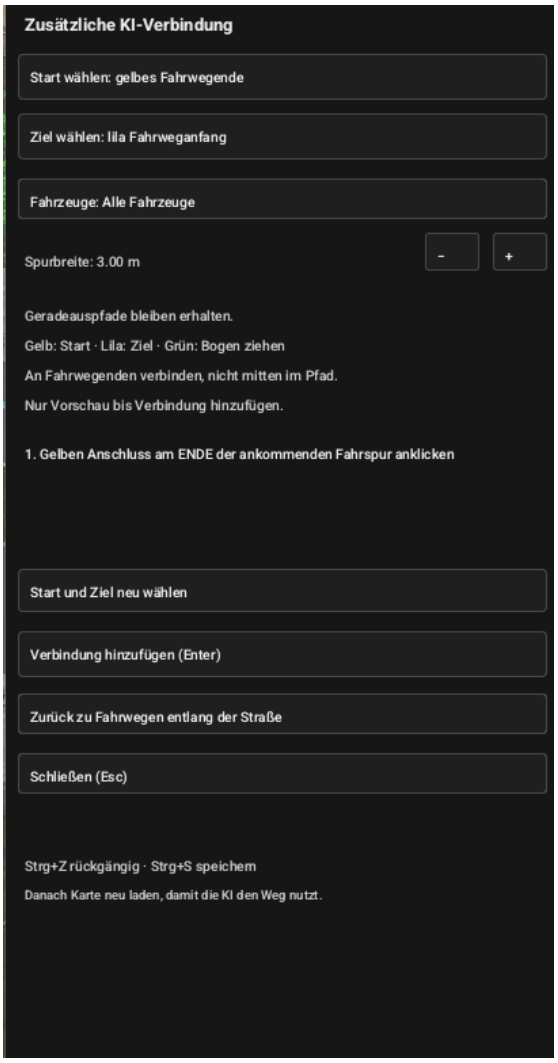
Normal dragging without freezes was confirmed on Linux with 0.7.20. The path display builds in the background. A brief map-update delay can still occur; diagnostics do not prove smooth operation on every map under all conditions.

Finish safely

Apply the preview, save with Ctrl+S, then reload the map for AI traffic. After troubleshooting, return to normal startup with start-editor.sh. Diagnostics stop; the improvements stay active.

23 Additional AI connection

Editor 0.7.22-pre



1 Open the tool

In the spline editor, open KI-Fahrwege entlang der Straße (AI paths along the road). Choose Zusätzliche KI-Verbindung erstellen (create an additional AI connection).

2 Choose start and target

Click the yellow path end of the incoming lane as the start. Then choose the purple path start of the desired outgoing lane as the target.

3 Shape the curve

Drag the green curve points while holding the left mouse button until the connection follows the desired route. Check vehicle permissions and lane width.

4 Add the connection

Click Verbindung hinzufügen (add connection), or press Enter. Until then, the connection is only a preview.

5 Save and reload

Press Ctrl+S, then reload the map so AI traffic uses the new connection.

Important: Straight-ahead paths remain intact. The additional connection extends the existing path network. Connections are only possible at path endpoints, never in the middle of a path.

See also pages 19-22. Screenshot: German interface.